Quantara AI — Audited Data Preparation & Model Suite

Comprehensive Forensic Audit, Pipeline Architecture & Machine Learning Validation Report

Target Asset Class: National Stock Exchange (NSE) Equities	**Time Span:** 10 Years (2016-10-24 → 2026-10-23)
Total Raw Observations: 19,776 Sessions (8 Stocks)	**Integrity Status:** 75/75 Checks Passed (SHA-256 Verified)

1. Executive Overview & Philosophy

This project establishes an institutional-grade machine learning data preparation and validation pipeline rebuilt from historical NSE equity data. Unlike standard academic demonstrations that suffer from lookahead bias or arbitrary price memorization, this repository enforces strict data stationarity, cryptographic tracking, and leak-free chronological boundary purging across **8 core Indian stocks**: *BHARTIARTL, HDFCBANK, ICICIBANK, INFY, ITC, RELIANCE, SBIN, and TCS*.

Three Core Governing Rules:

- **Zero Synthetic Data:** Not a single price bar has been hallucinated, interpolated, or artificially generated.
- **Strict Stationarity:** Raw price levels (e.g. ■2,500) are eliminated from inputs; all features are converted to dimensionless ratios.
- **Boundary Purging:** Samples whose forward prediction window overlaps subsequent splits are quarantined to guarantee zero lookahead leakage.

2. Phase 1 Forensic Audit: Correcting the Baseline

A forensic review of the original uploaded snapshot (source_snapshot/) revealed 4 fatal quantitative design flaws, which were completely corrected in the audited pipeline (data/splits/):

Design Dimension	Original Baseline Flaw	Audited Pipeline Correction
Model Features	Fed raw prices (Open, High, Low, Close, Bollinger Bands) directly into models.	Engineered **20 stationary features** (returns, moving average divergences, RSI, MACD %, volume differences).
Dataset Splits	Only 2 splits (80% Train, 20% Test); no validation set.	**3 Chronological Splits:** 80% Train (to Oct 2024), 10% Validation (to Oct 2025), 10% Test (to Oct 2026).
Boundary Leakage	No purging. Late training rows predicted returns overlapping into the test period.	**Purged boundary rows:** 16 (1d), 80 (5d), and 320 (20d) overlapping rows quarantined into purged_boundary_rows.csv.
Prediction Horizons	Hardcoded to a single 1-day timeframe.	Three distinct timeframes: **1-Day** (daily), **5-Day** (weekly), and **20-Day** (monthly).

3. Phase 2 Machine Learning Suite (5-Day Horizon)

Focusing on the **5-Day Forward Return Horizon (`horizon_5d`)**, four distinct model architectures were trained and benchmarked side-by-side on the untouched Test Split (**1,944 out-of-sample trading sessions** from late 2025 to late 2026). The classification target has 3 classes: **`DOWN`** (< -1.0%), **`HOLD`** ([-1%, +1%]), and **`UP`** (> +1.0%).

Model Architecture	Val Bal Acc	Test Accuracy	Test Bal Acc	Macro F1	DOWN Prec.	UP Prec.
0. Dummy Baseline (Chance)	33.33%	31.48%	33.33%	0.1596	0.0%	31.5%
1. Logistic Regression	34.69%	**37.65%**	**40.73%**	0.3557	**55.1%**	**36.3%**
2. Random Forest (100 Trees)	34.18%	37.09%	38.56%	**0.3687**	53.6%	32.0%
3. LightGBM (Tuned Trees)	**34.71%**	33.80%	34.97%	0.3374	45.8%	29.4%

4. Financial Strategy Simulation (Backtest with Real Friction)

A quantitative trading simulation was executed across all 1,944 test sessions, applying realistic Indian market round-trip friction (**0.15% transaction cost** for brokerage, STT tax, and exchange slippage):

Strategy Dimension	Buy-and-Hold Market Benchmark	Logistic Regression Model	LightGBM Classifier
Underlying Market Return	-0.29% gross (-0.44% net)	-0.29% gross (-0.44% net)	-0.29% gross (-0.44% net)
Long Trades Taken (`UP`)	1,944 sessions	791 trades	756 trades
Long Strategy Net Return	-0.44%	**-0.20% (+0.25% Net Alpha)**	-0.53% (-0.09% Net Alpha)
Downside Short/Hedge Signals	0 (Always long)	176 signals (**64.2% drop rate**)	415 signals (**57.1% drop rate**)
Downside Short Net Profit	N/A	**+0.79% Net Profit per Trade**	**+0.25% Net Profit per Trade**

5. Forensic Diagnosis: Why Is Model Accuracy ~38%?

A common beginner misconception is expecting 80% to 90% accuracy in stock forecasting. In institutional finance, market moves contain heavy noise. **38% balanced accuracy in a 3-class problem represents a true +5% edge over random guessing (33.3%)**. The performance ceiling in this dataset is dictated by four structural realities:

1. Severe Market Regime Shift: The training split (2017–2024) was a 7-year bull market (41% UP). The test split (2025–2026) was a market correction (40% DOWN). Tree models memorized bull breakout patterns that turned into bull traps during the correction.

2. Information Starvation: The model relies exclusively on 20 basic technical indicators. It is blind to the NIFTY 50 benchmark index trend, RBI interest rates, oil prices, quarterly corporate earnings, and macroeconomic news.

3. The Choppy 'HOLD' Trap: Small fractional moves (+0.95% vs +1.05%) arbitrarily switch labels between HOLD and UP, forcing the model to expend capacity trying to separate random daily market noise.

4. Stock Volatility Mismatch: A flat ±1.0% threshold applies equally to low-volatility utility stocks (TCS) and high-beta cyclicals (SBIN), diluting signal uniformity across sectors.

6. Feature Importance & Stock-by-Stock Alpha

Top Technical Predictors:

1. **volatility_20d_annualized:** Regime gauge (quiet vs wild).
2. **intraday_range_pct:** Candle bar expansion precedes swings.
3. **bollinger_width_pct:** Volatility squeeze indicator.
4. **macd_histogram_pct:** Momentum acceleration.
5. **macd_pct:** Medium-term trend direction.

Least Useful: Daily 1-day noise (daily_return_1d).

Stock-Specific Alpha (Test Set):

- **TCS.NS: 41.56% Accuracy**, **+0.57% Alpha** (Best)
- **INFY.NS: 35.80% Accuracy**, **+0.26% Alpha** (Strong)
- **BHARTIARTL.NS:** 30.45% Accuracy, +0.10% Alpha
- **HDFCBANK.NS:** 30.04% Accuracy, +0.07% Alpha
- **RELIANCE.NS:** 30.45% Accuracy, -0.40% Alpha
- **ITC.NS:** 34.16% Accuracy, -0.50% Alpha (Weakest)

7. Project Deliverables & User Tools

All trained models, evaluation metrics, and interactive tools are saved and fully functional in your project directory:

Deliverable Artifact	Location Path	Description / Function
Best Linear Model	models/horizon_5d/logistic_regression_baseline.joblib	Standardized multinomial classifier. Highest balanced accuracy (40.7%) and short profit (+0.79%).
Tree Model	models/horizon_5d/lightgbm_5d_classifier.joblib	Tuned LightGBM classifier with early stopping. Strongest on TCS (41.6% accuracy).
Interactive CLI Tool	scripts/predict_sample.py	Run live predictions on any stock: python scripts/predict_sample.py --symbol TCS.NS
Audit Reports	models/horizon_5d/phase2_complete_metrics.json	Complete machine-readable metrics, confusion matrices, and financial simulations.

8. Actionable Recommendations for Phase 3

Three High-Impact Pathways to Increase Predictive Power:

1. **Confidence Filtering (Trade Selectivity):** Instead of trading every session, require the model to have >50% predicted probability before triggering an entry. This eliminates noisy low-conviction trades.
2. **Binary Target Formulation (Drop HOLD):** Reframe the prediction target to binary direction (`UP` vs `DOWN`) or large momentum breakouts (> ±2.0%), eliminating the noisy middle zone.
3. **Multi-Horizon Models:** Expand this architecture to the **20-Day Horizon (`horizon_20d`)** where macroeconomic and trend signals typically have significantly higher signal-to-noise ratios than weekly fluctuations.

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